Motion Station Final Report

Executive Summary

The Boston Children's Museum requested that our team of Northeastern students design and construct an interactive exhibit for visitors ages 4–8 that introduces mechanical engineering concepts while incorporating themes of European culture and the Pixar film *Cars*. The exhibit needed to be safe, inclusive, and engaging while operating within a $120 budget and a three-month design timeline. Our team designed and built an Italy-themed game board where children can assemble different sized gears along the side of the board and turn them to drive a pulley system that moves a toy car along a straight racetrack path. The gear-and-pulley mechanism teaches the mechanical engineering principle of gear ratios through hands-on exploration, requiring no reading or complex instructions. The board features 9 interchangeable gears of 3 different sizes, allowing children to experiment with how gear size affects the car's speed and the effort required to move the system. Lights and speakers provide sensory feedback throughout the experience. The design was analyzed using specifications determined from the desired functions and traits. Notable results include an average playing time of 102 seconds, 3.85 second average car reset time, and an average of 6 gears used per player. The cost of the project reached $111.03 of the $120 budget. Possible future improvements include increasing electronic components, adding more cars and pathways, and refining the gears to fit together more seamlessly.

Introduction

The Boston Children's Museum has requested the design and construction of an interactive exhibit with multiple installments connected by a common theme. Each section of GE 1502 has selected a different theme. This report focuses on one section's chosen theme of "Day in the Life of an Engineer" combined with global exploration and Disney movies. Each group in the class section has designed a game based on one engineering discipline, one continent and one Disney movie. Team Motion Station has selected Mechanical Engineering, the continent of Europe and the Pixar film *Cars.* The challenge for this team is to incorporate all of these themes in a way that is easy for the children to comprehend while not seeming incohesive.

The children who will play in this exhibit are young, curious, and in early developmental stages. Children in this age range (4-8 years) learn best through exploration, play, and hands-on experience rather than observation or textual instructions. The exhibit must communicate concepts through physical interaction and visual cues rather than relying on written instructions or complicated rules.

The museum's educational philosophy seeks to spread 8 Habits of Mind to users. The specific exhibit addressed by this memo focuses on habit number 7, "Understanding Different Perspectives". This habit encourages children to recognize that people in different places, cultures, and professions experience the world in various ways. The children will learn about

many parts of the world and types of engineering at the same time. The Disney themes will facilitate the children's ability to contextualize their connected global differences and themes. It is important that all types of children are exposed to engineering disciplines as well, especially girls, in order to spark their curiosity about these topics from a young age and work to bridge the gender gap in the STEM fields.

Background Information

Children and Gender Roles

One important consideration in the design of this project was in how it might affect the children's perceived gender roles, especially due to its cars and mechanical engineering themes which are typically regarded as more masculine. We researched the relationship between children's games and gender roles in order to better understand how to work to eliminate this stereotype. One Chinese study found that children reinforce traditional gender norms in their play through clothing, appearance, language patterns, and gender-typed roles (boys as authority figures, girls as nurturers)[4]. A study in Ireland which tested students' persistence in a scientific experiment-based game found that boys persisted for longer than girls in this game, suggesting that male domination in STEM fields leads to an increased confidence in boys even from a young age [5]. A third observational study demonstrated that children are more likely to choose partners of the same gender in games, specifically boys when it comes to competitive or athletic games. This pattern can be reversed by creating a game that encourages teamwork across diverse users [9].

Children and Technology

Because this game includes technology, we also conducted research on the effects of technology on children in order to ensure that the technology employed in our exhibit was effective and safe for the children's development. The studies demonstrated a variety of results. The first study found that social skills were significantly associated with daily internet use time and frequency of traditional game play, and that too much screen time can negatively impact a child's social skills [1]. A second study, however, found that digital technologies have the potential to expand young children's experiences and skills upon observing a cohort of pre-schoolers and their behaviors with digital technology [2]. This duality is important to note and demonstrates that a balance must be found between those technology uses that aid and harm development.

Gear Ratios

Gears can exist in any size, and the gear ratio, or ratio between the sizes of any two gears, can allow us to calculate the speed at which they will turn together. The number of teeth on a gear is equal to its diameter multiplied by 8. When two gears are spun together, one is called the "drive gear" and the other is called the "driven gear". The "gear ratio" of these can be defined as the ratio between the diameter of the driven gear: diameter of the drive gear. This ratio directly relates to the speed at which the gears turn. This means, for example, that if the diameter of the

driven gear is three times as large as the diameter of the drive gear, then the drive gear will spin three times as fast [6].

Code

The electronics in this project were coded in Python and powered through a Raspberry Pi Pico microcontroller board. The code relies on modules built into MicroPython, which is a lightweight version of Python designed for microcontrollers. These built-in modules provide pre-written functions that handle complex hardware communication, so the programmer can easily set up an interface with the electronic components. For example, the I2S audio module handles the process of sending digital sound data to the amplifier. The amplifier then converts these digital signals into electrical signals that vibrate the speaker to produce sound. The digital audio data is generated using trigonometry, specifically sine waves, which mathematically describe the motion of the speaker cone [7]. The documentation for the module, which was used in the code, can be found in the works cited page below [8]. The break beam sensors use their own separate code library, which is a set of commands written and published by other developers to make hardware interactions more accessible.

Problem Definition

The final project was displayed and used on April 12, 2026 by a cohort of museum visitors accompanied by their parents. The client, the Boston Children's Museum, requested the design and construction of an interactive exhibit for the users, children mostly aged 4-8 who learn best through hands-on exploration. The museum attendees are primarily children and families from the Boston area. This project attempts to address various needs of the children, including that of having fun, of an accessible game that can be adapted to any child, and of a game that educates based on exploration rather than rules. The design team consists of four engineering students with basic electronics and coding knowledge, access to makerspace facilities, and a budget constraint of $120 maximum for all materials. The Boston Children's Museum seeks to spread its 8 Habits of Mind to the users who are in early developmental stages. In this specific exhibit, our team focused on addressing habit number 7, "Understanding Different Perspectives" by building a system where kids can learn about how different sizes of gears affect movement, an important mechanics aspect, using the themes of Italy and the Pixar film Cars.

Additional stakeholders of the project include the parents of the children and Northeastern university. The parents of the children will be accompanying them to the museum, so they will also be interacting with the game, or at the very least watching their children interact with it. As parents, they care about the education, safety, and satisfaction of their children. Northeastern is a stakeholder as the engineers represent the university's name. Any work presented at the museum is tied to the university, meaning that the students have been entrusted with upholding the reputation of the university.

The project assignment also includes some specific constraints. The project must implement an interactive electronic component, specifically using a Raspberry Pi Pico to manage electronic elements which will help enhance the gaming experience. The design is supposed to be free of latex and edible materials. The game must be designed and fabricated within a three-month timeframe as part of the GE 1502 course requirements, with the final installation scheduled for April 12, 2026. These resource limitations mean that the solution must be achievable with relatively simple materials while still delivering a unique and engaging experience that justifies the museum's investment in requesting this exhibit.

The following section specifies a problem definition for the specific solution of team Motion Station, demonstrating how the team decided to address the needs of the children and museum.

Functions

As a team, we have determined that successful solutions must include the following functions:

- **Car moves across track:** The car must move back and forth across the racecar track in order for the game to work successfully and provide entertainment and satisfaction. When the car moves, the child will be rewarded for their effort and therefore able to understand that they have succeeded without specific instructions.
- **Gears are moveable:** The gears can be easily moved and rearranged by the user in different patterns, allowing for exploration. The user can learn from trial and error.
- **Gear size affects car speed:** The gears chosen by the user directly affect the difficulty and therefore speed at which the mechanism moves, teaching the user about gear ratios through experience.
- **Movement directly caused by user input:** Changes in the user's movement visibly affect the movement in the mechanism, allowing the user to see the results of their choices and actions in real time.
- **Detect when car passes threshold:** A sensor placed at a certain point of the car's path must accurately sense when the car passes by. Creates a clear sign of success for the user, who might not otherwise understand that the goal has been reached.
- **Lights and speaker triggered by break beam sensor:** When a car is detected by the sensor, a response of lights and sound must initialize. Sensory feedback makes the exhibit accessible to children of a wide range of ages, language abilities, and educational level.

Constraints

- **Limited time for playing:** On the day of the final presentation, the children will be traveling through many different games and even different exhibits. They will likely not remain at each station for more than 3-5 minutes.
- **Limited time for design:** The design team is working within a 3-month time constraint as the exhibition is scheduled for April 12, 2026.
- **Materials**: Certain materials including latex and anything edible are prohibited by the museum. These rules are in place to ensure the safety of the users.

- **Money:** Total material costs cannot exceed $120, because the design team consists of college students. The budget must cover all physical components, electronic components, fabrication materials, and any necessary tools or supplies not already available in the maker space.
- **Fine motor skills:** An important factor which must be considered in the design process is the limited fine motor skills of the users. Children ages 4-8 are still developing hand-eye coordination, grip strength, and precise movements, so the interactive elements of the exhibit must be designed to be easily moved and handled.
- **No reading requirement:** Many of the children who will be playing the game might not have learned how to read in school yet, therefore they cannot be expected to read any instructions. This means that there must be alternative pathways for understanding the flow of the game such as vocal or visual cues.
- **Electronic component:** The project guidelines require the implementation of an interactive electronic component in this exhibit. This can be achieved by using buttons, lights, sound etc.

Most Important Traits

Solutions should exhibit these 6 traits, ranked in order of importance:

1. **Safe:** Safety is an extremely important consideration for this design. Young children are not equipped with a complete understanding of danger. Because of their curiosity, they might put objects into their mouths or act in other unpredictable ways. Thus extra care must be put into the exhibit's safety.
2. **Inclusive:** Especially in early developmental stages, it is crucial that children learn to include and to be included. A game must clearly communicate openness to all. In the context of this exhibit, inclusivity includes considerations about accessibility, mental, emotional and physical development stages, language barriers, and stereotypes such as gender and race.
    a. **Sensory feedback:** because most children in this age range cannot read, the exhibit must include ways for the children to understand the games and explorations using elements such as images, color, and sound rather than text.
3. **Exploratory:** rather than a challenge with a winner or one "right" answer, an effective exhibit should allow each child to discover new ideas through their own path. It is important for children to learn through trial and error and observing various outcomes for their actions during the developmental stages.
4. **Fun:** The children's main motivation for coming to the museum will be to have fun. Appealing to the user's needs and desires is an important part of the design process. If the kids are having fun they will be more likely to engage further with the game and participate in the educational aspects as well.
5. **Educational:** one purpose of this exhibit is to introduce children to mechanical engineering and widen their global perspectives. Because the children are young, the

game must effectively convey the essence of mechanical engineering disciplines in a basic and straightforward way.

   a. **Mechanical engineering principles:** at least one specific mechanical engineering principle should be demonstrated, such as one of the basic machines or very basic physical concepts, in order for the children to gain a true understanding of what this field involves.

6. **Visually appealing:** children will be significantly more likely to engage with a game if it looks good. Because they are in the process of understanding the world around them, objects with exciting or recognizable features will increase engagement.

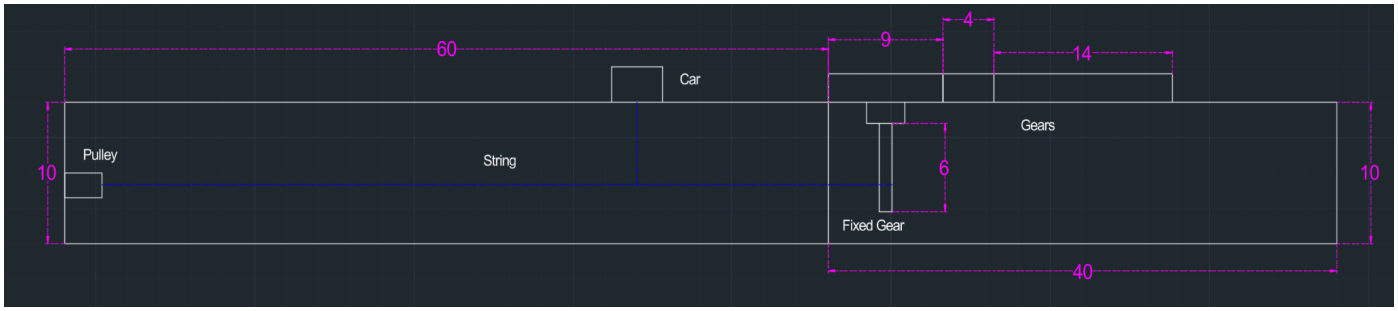
	Fun	**Exploratory**	**Educational**	**Inclusive**	**Safe**	**Visually Appealing**	**Total**
Fun	X	0	1	0	0	1	2
Exploratory	1	X	1	0	0	1	3
Educational	0	0	X	0	0	1	1
Inclusive	1	1	1	X	0	1	4
Safe	1	1	1	1	X	1	5
Visually Appealing	0	0	0	0	0	X	0

Description of the final Design

Our final design consists of a laser-cut box (30 x 60 x 10 cm) with Italy-themed images on its surface as well as designs from the Cars movie. A straight cutout “racecar track” runs vertically across the lid of the box, with a toy car attached to a string-and-pulley system that moves underneath this track. Attached to the track is another laser-cut box (40 x 20 x 10 cm) containing 3 laser-cut spur gears of three different sizes (4, 9, and 14 cm respectively) with 9 gears total. The top of the second box has a slot in the end closest to the first box which holds a fixed gear on a rod. Next to this slot, there is a line of velcro along the rest of the box on which the user can place additional gears of their choice. The gears are inserted into pegs with velcro at the bottom in order to be easily rearranged. Because the gear options are different sizes, the velcro makes the rods moveable in order for any gear combination to fit properly. Once the gears are placed, the first gear can be turned, which drives the second gear and continues all the way to rotate the fixed gear. Both boxes have holes in the sides through which the string passes. The rod underneath the fixed gear connects to and rotates a twine string which passes into the first box through the hole. The string runs up the length of the racetrack in the first box, looping through a pulley at the end. A toy car is connected vertically to the string, and moves along the top of the track as the bottom string rotates. Inside the box, a Raspberry pi pico is placed and connected to

a LED strip, a speaker and amplifier, and a break beam sensor through a bread board. Based on the sizes and ratios of the chosen gears, the pulley-rope system will move at different speeds, allowing the children to experiment with the ratios.

Figure 1: Side View of Design Showing Gear-and-Pulley System (Dimensions in cm)



The gear and pulley system shown in the image above allows the children to move the cars by cranking the gears. The black rectangle on the left represents the laser-cut box which houses the electronics and pulley system, and the black square on the right represents the box which houses the gears that can be interchanged by the user. The leftmost gear is the one that remains fixed at the top of the box, as it is attached to an axle that goes down into the box. Users can place and spin their own gears. The gears on the right spin the fixed gear on the left, and the axle at the bottom spins as well. A string is attached around the axle, and around a pulley on the opposite end of the box. The spinning motion causes the rope to move in a closed loop, which moves the car. The car is attached to the string via a second, vertical string. When the car passes the break beam sensor, the vertical string breaks the beam, which sets off a rainbow pattern on the LED light strip and the theme to “Life is a Highway” on the speaker.

Typical gameplay goes as follows: a child approaches the exhibit and sees the Italy-themed racetrack. They place the gears into their rods and along the central velcro strip in any distribution of their choice. Once they have set up their gears, they can spin the gear on the farthest end and watch the car move as they spin. Once the car passes the break beam sensor (near the end of the track), rainbow lights flash and the theme from “Life is a Highway” plays. This interactive design will allow for the children to see in real time how their choices affect the outcome of the car’s speed.

Figure 2a: Pulley SOLIDWORKS Design

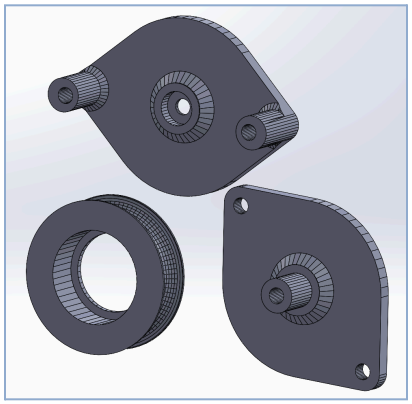


Figure 2b: Pulley Final Version



Figure 3: Gear Holders SOLIDWORKS Design

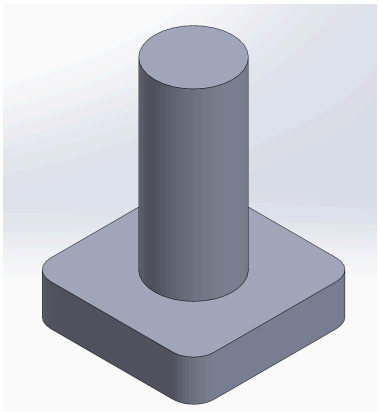
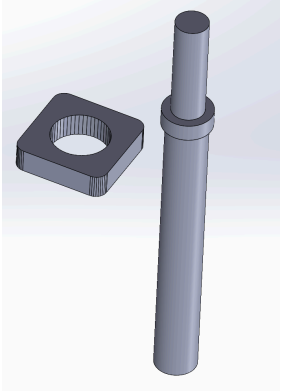


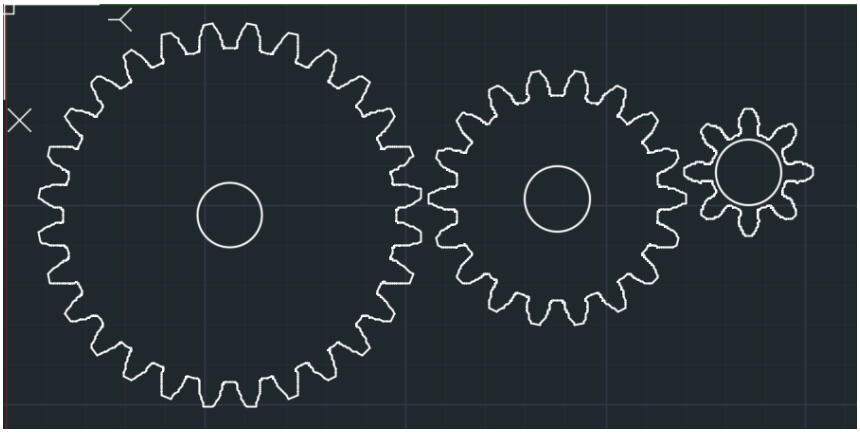
Figure 4: Rod for Fixed Gear SOLIDWORKS Design



Description of mechanical parts

Each piece was designed in SolidWorks, then sliced in Cura, and 3D-printed using PLA plastic filament. The pulley was designed to fit a 608 bearing as a way to reduce friction when in use. The gear holder peg has a diameter of 2.45 cm, and a base of 4 x 4 x 1 cm. The rod has 3 diameters: 2.45 cm, 3.5 cm, and 3 cm (from top to bottom in Figure 4), and a square spacer ring with a diameter of 3.05 cm, and size 4 x 4 x 1 cm. The system operates as follows: the gear holders (Figure 3) are set in place with gears attached to them. The driven gear is mounted onto the rotating rod (Figure 4) using the square spacer ring which ensures the gears are all properly aligned. As the gears rotate, the driven gear then spins the rod, translating the gears rotational motion into linear motion. A string is attached to the rod and the pulley (Figure 2), and the pulley allows the string to move with minimal friction.

Figure 5: Autocad Design For Gears



The gears were designed in AutoCAD and 3D printed in 0.25 inch plywood. They were designed using a Gear CAD Generator website cited below (GearGenerator.com). The intended design was for the gears to be 4, 8 and 12 cm in diameter in order to have a 1:2:3 ratio. The final gears resulted in a 4:9:14 ratio. This is likely due to scaling error between the CAD and laser cutter.

Description of code

The code mechanism will run as follows: after initial hardware setup, the program enters a loop in which the break beam sensor is on and checks for a break in the beams every 50 milliseconds. When the beam is broken, meaning that a toy car has passed across the top of the track, the program opens the pre-loaded "Life is a Highway" WAV file and streams the audio data to the amplifier via the I2S protocol. The LED strip also cycles through a rainbow pattern, with the lights updating every 8th audio chunk to avoid overwhelming the Pico. Once the song finishes, the LEDs clear and the program returns to monitoring the sensor, waiting for the next car to pass.

Figure 5 shows a flow chart for the code. Raw code can be found in Appendix C below.

Figure 5: Flow Chart for Code

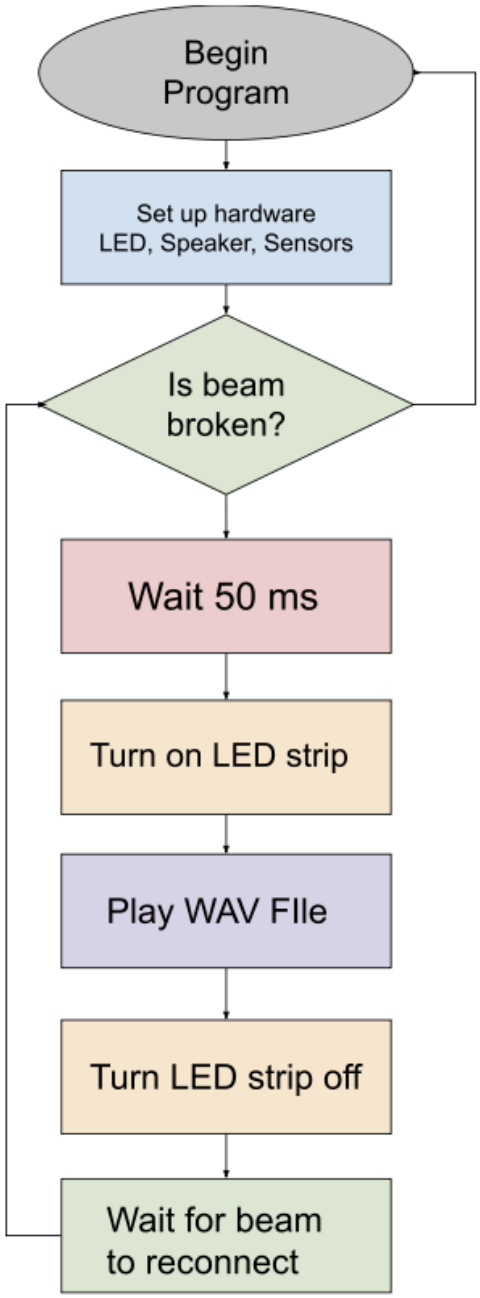


Figure 6: Bread Board with LED Strip and Speaker Wiring

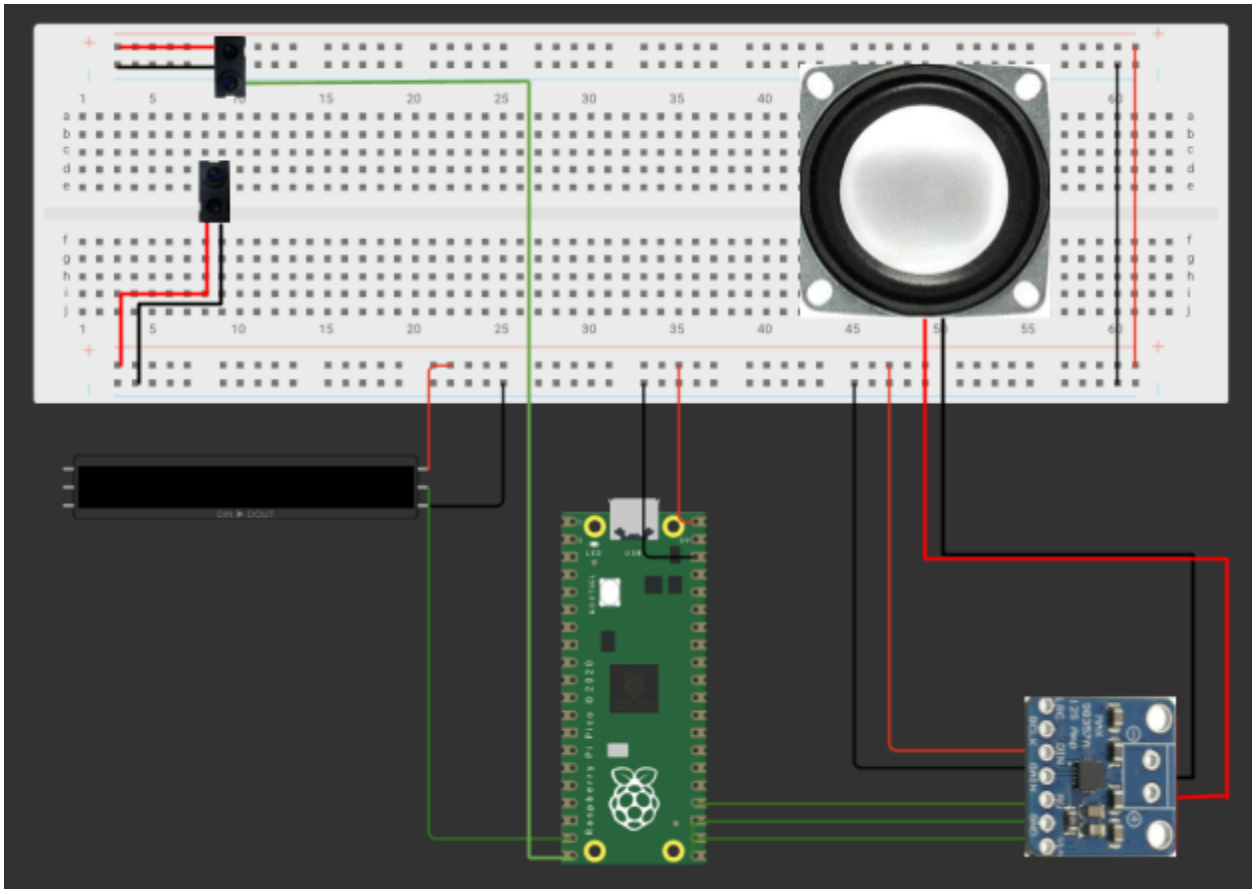


Figure 6 demonstrates the Raspberry pi pico board with respective wiring for the speaker-amp sound system, LED strip and break beam sensor. The speaker is a EK1725 module that connects to the Raspberry pi through a MAX98357A module IS2 class D Amplifier.

Analysis

Budget analysis: The total cost of this project reached $111.03, which is $8.97 less than the budget of $120 dollars. This cost averages out to $27.76 per person.

Function/Trait	**Spec**	**Goal**	**Actual**
Car moves on track	String moves car from one extremity of track to the other	100% of times	100% of times
	Car reset time	< 4 seconds	3.85 seconds
Moveable Gears	Time to reset gears	< 10 seconds	5 seconds
Gear Size Affects Speed	Number of distinct speed outcomes	3	3

	Ratio between largest and smallest gear combination	1:3	1:3.5
Input Causes Movement	% of times gear turns caused car movement	80	100
Senses Car Passing	% of accurate response to beam break	100	70
Sensor Triggers Electronics	% of successful response to beam break	100	100
Accessible to Children	Crank handle diameter	> 2 cm	2.45 cm
Engaging	Average engagement duration	90 seconds	102 seconds
	Number of gears used	3	6
Educational	Number of engineering principles	2	2
Safe	Smallest gear diameter	> 2 cm	4 cm

Based on the results above, most of the team's design goals were either met or exceeded during the exhibition on April 12. The average time for car reset resulted in 3.85 seconds, just under the goal of 4 seconds, and gear resets took approximately 5 seconds, significantly under the 10-second goal. The three gear sizes produced three visibly different speed outcomes as intended, and the ratio between the largest and smallest possible gear combinations reached 4:14 (roughly 1:3.5), slightly exceeding the target ratio of 1:3 but still giving children a noticeable difference in car speed for different combinations. This slightly different ratio was likely caused by imperfections in scaling with the laser cutter and AutoCad design file. Engagement exceeded expectations, with an average of 102 seconds spent at the exhibit against the goal of 90 seconds. Each player used an average of 6 gears per session, which is twice as many as the initial target of 3. This high number of different gears suggests that the design's exploratory aspect encouraged the children to learn by experimentation. Additionally, 100% of gear turns led to the car's movement, demonstrating a visible connection between user input and output.

We also made some qualitative observations at the museum. First, many children were aided or prompted in some way by their parents. The most commonly chosen gear was the blue gear, likely because it was the largest and easiest one to handle. The most common mistake in gameplay was the children attempting to move the pegs rather than the gears themselves. This mistake is due to the fact that there were no written instructions. Many children showed visible excitement when the car reached the end of the track and audio played. Finally, some children stacked the gears on top of one another rather than side to side. While this move did not affect the gear ratio, it actually helped stabilize the mechanism as a whole. This discovery highlights the benefits of exploratory play.

The primary shortfall came from the break beam sensor, which registered only 70% of car passes against the 100% goal. When the sensor did register a pass, however, it triggered the lights and speaker successfully 100% of the time, indicating that the issue lies in sensor detection. Rather than bugs in the code or shortcoming of electronics, this error was likely due to the sensitivity of the break beam sensor, which tends to produce faulty connections if the two components are not exactly in line with one another. Because the children were frequently moving around the box and table, the sensors were likely moved out of line as well. Safety and accessibility specs were all met as the smallest gear remained above 1.75 inches to prevent choking risk. Two mechanical engineering principles (gear ratios and pulley systems) were successfully demonstrated, meeting the educational goal.

Conclusion

In summary, Team Motion Station has developed a final product of an interactive exhibit for the Boston Children's Museum. By using the themes of Disney' s *Cars*, Italy, and gear mechanics, we created a hands-on experience for children from 4-8 years old to explore complex mechanical engineering concepts like gear ratios without needing much instructions. As a team, we prioritized creating an inclusive and safe environment so every kid who comes to our station can have fun and learn something new.

The final product is an Italy-themed laser-cut game with nine laser-cut gears in three sizes, which children arrange on a velcro strip to drive a pulley system that moves a toy car along a track. When the car reaches the end of the track, a break beam sensor triggers a rainbow LED pattern and the "Life is a Highway" theme. Testing on April 12, 2026 showed that the exhibit met a majority of its design targets. Children engaged with the exhibit for an average of 102 seconds and used an average of 6 gears per session, both above goal, while the gear-and-pulley mechanism produced three distinct speed outcomes and a ratio range slightly larger than the target of 1:3. Safety, accessibility, and educational goals were fully met, as the game demonstrated two mechanical engineering principles and all components were sized correctly for young children. The main shortfall was the break beam sensor, which only detected 70% of car passes, though the electronic response was accurate when triggered. The total project cost was

$111.03, within the $120 budget. Overall, the exhibit successfully combined its cultural and engineering themes into an exploration that allowed children to learn about gear ratios through trial and error rather than instructions, catering to the museum's educational mission and the goals established in the problem definition. Recommendations for future iterations can be found in the following section.

Future Recommendations

While we met and even exceeded some of our goals, there are aspects of the game that can be improved to elevate the experience of the user. One important aspect of the exhibit that could be improved could be the strength of the components. We predict eventual wear and tear of the exhibit's individual elements due to constant interaction by the children. The gears could be redesigned to fit together more seamlessly and within the original plan for a 1:3 ratio. Another possible addition for the future would be to expand the possibilities for exploration and replayability by adding more gear sizes to increase the number of possible combinations. The game's educational and aesthetic scope could be expanded by making the board interchangeable so the map surface could be swapped out for other countries. Additionally, the implementation of more electronic components, such as a screen, buttons to track laps or reset, or more sounds from the speaker, would allow for more sensory interaction. Another possible improvement is adding more cars and pathways to increase both exploratory play and the amount of users that can play the game at once. Finally, the design could continue to be tested with more children over a longer period rather than just one afternoon in order to gain more accurate specs for further improvement.

Appendix A: Time Sheet

	Jonas	Milkias	Ludo	Katy	TOTAL HOURS	Major Accomplishment of the Week
Week 1	0	1.5	1.5	0	3	Creating and submitting team charter. Initial brainstorming.
Week 2	1.25	1.25	1.25	1.25	5	C&T Prototype and proposal
Week 3	1	1	1	1	4	First oral presentation
Week 4	0.5	0.75	0.75	0	2	Narrowing down idea using feedback from initial proposal & creating a basic design
Week 5	1.5	1.5	2	1.5	6.5	Milestone report outline, started work on all 4 physical components
Week 6	2	1	3	0	6	milestone report and presentation
Week 7	6	4	7	2	19	Built electronics, wrote code, laser cut box, 3D printed gears
Week 8	2	0	0	0	2	Reiterated major design components to kickstart new prototyping
Week 9	3	0	3	0	6	Began physical creation of updated design
Week 10	5.5	0	6	0	11.5	Prototyping and reiterating original designs
Week 11	16	3	19	2	40	Finalizing project for dry run
Week 12	6	5	10	3	24	Presenting project at Museum
Week 13	2	1	7	1	10	Writing Final Report and Presentation
Individual Sums	46.75	20	61.5	11.75	139	

Appendix B: Cost Sheet

Total spent	111.03	**Amount Remaining**	8.97		
Purchases	**Date**	**Amount ($)**	**Item Purchased**	**Purchased from where**	**Teammate who bought**
Purchase 1	3/1/26	21.99	Toy Cars	Amazon	Ludo
Purchase 2	3/1/26	11.49	Speaker	Amazon	Ludo
Purchase 3	3/1/26	6.88	Amp	Amazon	Ludo
Purchase 4	3/9/26	25.29	Wood	EXP	Ludo
Purchase 5	3/28/26	5.95	Wood	EXP	Ludo
Purchase 6	3/30/26	9.44	Bearings	Grainger Ind. Supply	Jonas
Purchase 7	3/30/26	3.44	New Amp	Amazon	Ludo
Purchase 8	4/11/26	26.55	Plastic Bin	Target	Jonas

Appendix C: Raw Code

```
#-----------------------------------------------#
# sound_beam_leds.py                            #
# by Ludo                                       #
# v2.1,  03/28/2026                             #
# Plays "Life is a Highway" and flashes         #
# LED strip when break beam sensor is triggered  #
#-----------------------------------------------#

import struct
import time
import machine
import neopixel # for LED
from machine import Pin, I2S #Speaker Module

#Pin Definitions
# I2S Audio pins
SCK_PIN  = 16   # BCLK - clock
WS_PIN   = 17   # LRC - communicates with amp
SD_PIN   = 18   # DIN - carries audio data

# Break beam sensor (receiver side)
BEAM_PIN = 15

# LED strip
LED_PIN  = 14
NUM_LEDS = 8

# Setup Hardware
beam = Pin(BEAM_PIN, Pin.IN, Pin.PULL_UP)
old_beam_value = beam.value()

WAV_FILE = "highway_song.wav"

strip = neopixel.NeoPixel(machine.Pin(LED_PIN), NUM_LEDS)

# LED Functions
def clear():
    for i in range(NUM_LEDS):
        strip[i] = (0, 0, 0)
```

```
    strip.write()

BRIGHTNESS = 0.3  # 0.0-1.0, keep low to reduce power draw

def scale(val):
    return int(val * BRIGHTNESS)

#LED strip flashes a rainbow
def rainbow_step(offset):
    for i in range(NUM_LEDS):
        pos = (i * 256 // NUM_LEDS + offset) % 256
        if pos < 85:
            strip[i] = (scale(pos * 3), scale(255 - pos * 3), 0)
        elif pos < 170:
            pos -= 85
            strip[i] = (scale(255 - pos * 3), 0, scale(pos * 3))
        else:
            pos -= 170
            strip[i] = (0, scale(pos * 3), scale(255 - pos * 3))
    strip.write()

# Audio Helper Functions
# Reads sample rate and channel count from the WAV file header.
def read_wav_header(filename):
    with open(filename, "rb") as f:
        header = f.read(44)
    sample_rate  = struct.unpack_from("<I", header, 24)[0]
    num_channels = struct.unpack_from("<H", header, 22)[0]
    return sample_rate, num_channels

# Plays song and lights at once
def play_song_with_lights(audio):
    buf = bytearray(4096)
    rainbow_offset = 0

    f = open(WAV_FILE, "rb")
    f.seek(44)  # skip the WAV header

    # Need this to not overwhelm the pico with commands
    chunk_count = 0
```

```
    while True:
        num_read = f.readinto(buf)
        if num_read is None or num_read == 0:
            break

        audio.write(buf[:num_read])
        chunk_count += 1

        # only update LEDs every 8th chunk
        if chunk_count % 8 == 0:
            rainbow_step(rainbow_offset)
            rainbow_offset = (rainbow_offset + 3) % 256

    f.close()
    clear()

# Read WAV header and set up I2S audio
sample_rate, num_channels = read_wav_header(WAV_FILE)
fmt = I2S.MONO if num_channels == 1 else I2S.STEREO
print(f"Loaded {WAV_FILE} ({sample_rate}Hz, {num_channels}ch)")

audio = I2S(0, sck=Pin(SCK_PIN), ws=Pin(WS_PIN), sd=Pin(SD_PIN),
            mode=I2S.TX, bits=16, format=fmt,
            rate=sample_rate, ibuf=40000)

# Main Loop
print("Ready. Waiting for beam break...")

while True:
    current = beam.value()

    if current != old_beam_value:       # state changed
        old_beam_value = current
        if current == 1:                # beam just broke (went HIGH)
            print("Beam broken! Starting lights and sound...")
            play_song_with_lights(audio)
            print("Done. Waiting for next trigger...")

    time.sleep(0.05)
```

Appendix D: Original Design Images

Figure 7: Gears SOLIDWORKS Design

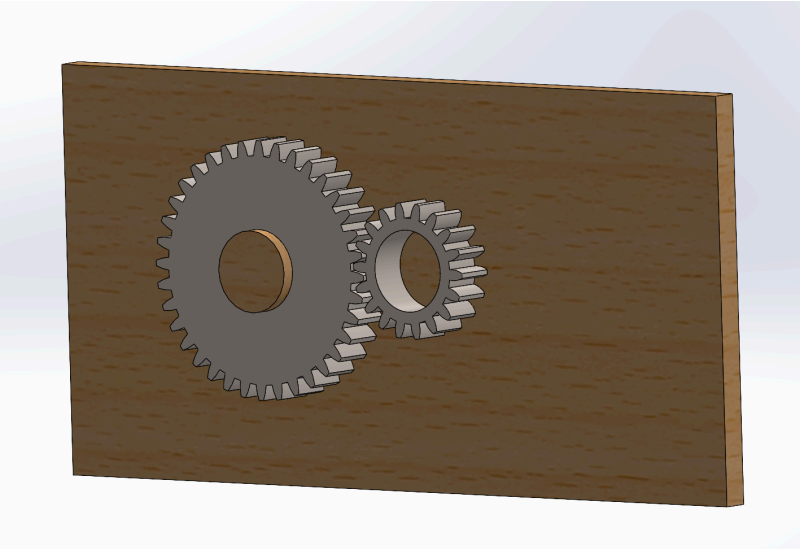
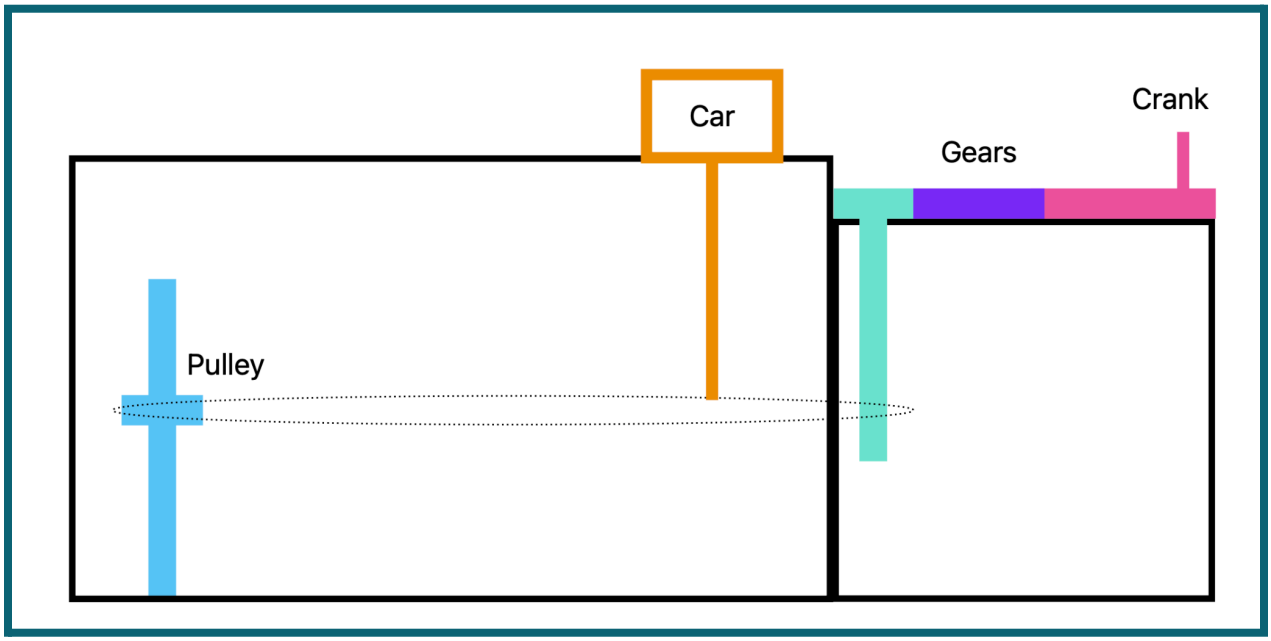


Figure 8: Preliminary Design



Appendix E: AutoCad Images

Figure 9: Autocad Design For Box

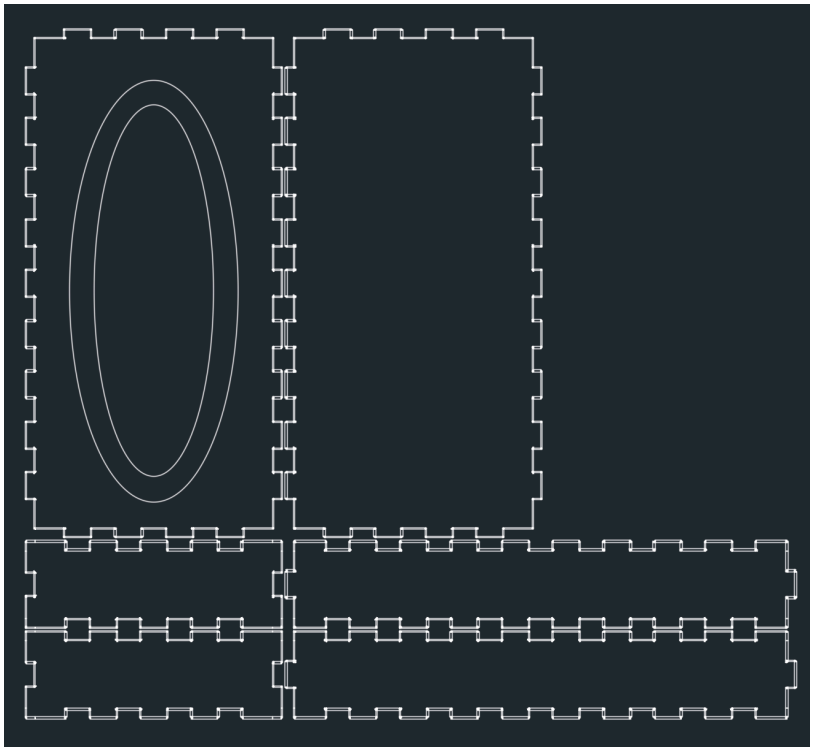
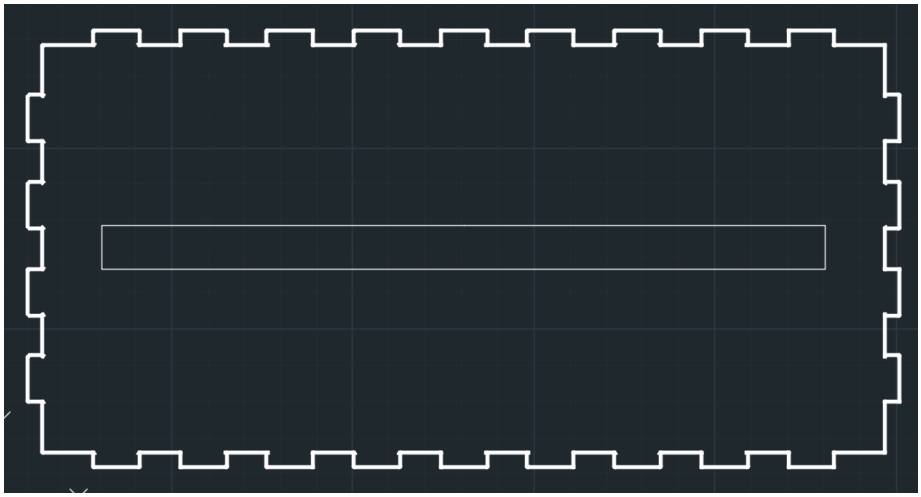


Figure 10: Autocad Design For Top of Box



‘

Figure 11: Autocad Design For Second Box

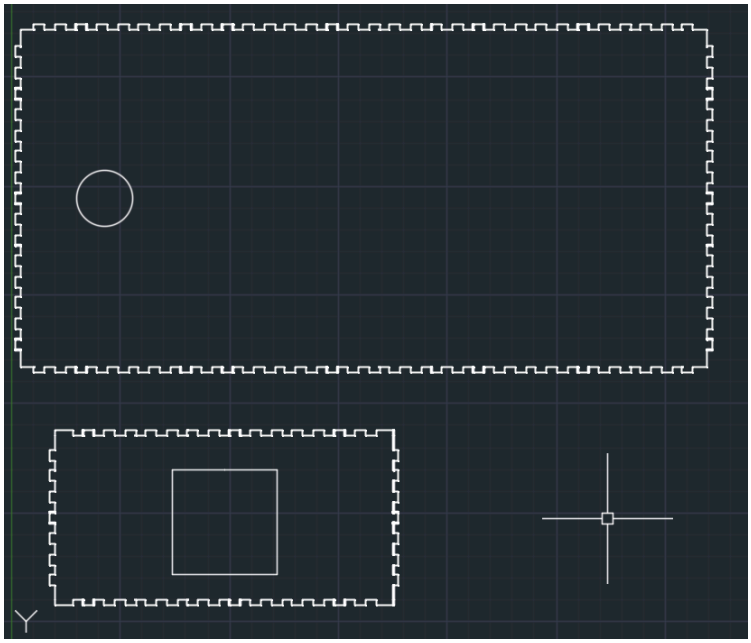
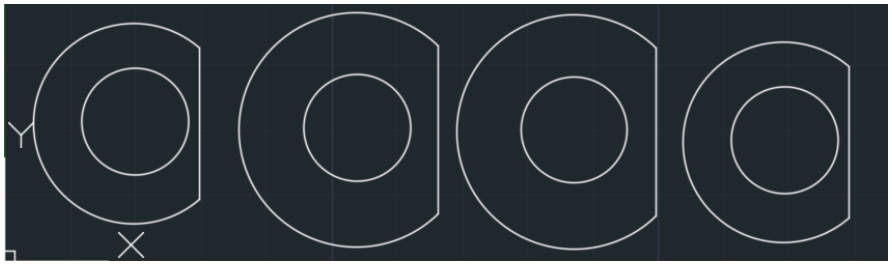


Figure 12: Autocad Design For Inner Supports



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